Program

```
#include <stdio.h>
#define MAX 50
int graph[MAX][MAX];
int visited[MAX];
int n;
void DFS(int vertex)
{
   int i;

   // Mark vertex as visited
```

```
   visited[vertex] = 1;
   // Display vertex
   printf("%d ", vertex);

   // Visit adjacent vertices
   for (i = 0; i < n; i++)
   {
      if (graph[vertex][i] == 1 && visited[i] == 0)
      {
         DFS(i);
      }
   }
}
int main()
{
   int i, j;
   int start;
   // Read number of vertices
   printf("Enter number of vertices: ");
   scanf("%d", &n);
   // Read adjacency matrix
   printf("Enter the adjacency matrix:\n");
   for (i = 0; i < n; i++)
   {
      for (j = 0; j < n; j++)
      {
         scanf("%d", &graph[i][j]);
      }
   }
   // Initialize visited array
   for (i = 0; i < n; i++)
   {
      visited[i] = 0;
   }
   // Read starting vertex
   printf("Enter starting vertex: ");
   scanf("%d", &start);
   printf("DFS Traversal: ");
   DFS(start);
   return 0;
}
```

Sample Input

Using the same graph as the BFS example:

Enter number of vertices: 5

Enter the adjacency matrix:

0 1 1 1 0

1 0 1 0 0

1 1 0 0 1

1 0 0 0 0

0 0 1 0 0

Enter starting vertex: 0

Output

DFS Traversal: 0 1 2 4 3